SO cross-cl pipeline: decision tree — Bayesian search + profile scheme

left: search for moments with the evidence (r marginalised) · right: profile scheme, every model at its own r

A · Bayesian search for moments (evidence, r free)

run BBPower without moments — $M_{0,r}$

is χ^2 good? (PTE > 0.01)
- No → **Res_wr**: residuals of $M_{0,r}$ all parameters fixed at their best fit (r too); test the d and s families, each with its own β free
- Yes → is r compatible with zero at 1σ?
  - No → **Res_wr**: residuals of $M_{0,r}$ all parameters fixed at their best fit (r too); test the d and s families, each with its own β free
  - Yes → upper limit

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moment detected? (*) 0 outside the 68% HPD of an amplitude
- Yes → $M_{1,r}$ = $M_{0,r}$ + a detected family, in order of residual evidence (**); full fit (all parameters free), pruned (***)
- No → **Res_r0**: residuals of $M_{0,0}$ (r = 0) all parameters fixed, own β free; test the families d, s, dT, d2, s2, ds1, ds2

$M_{1,r}$ = $M_{0,r}$ + a detected family, in order of residual evidence (**); full fit (all parameters free), pruned (***)

is $M_{1,r}$ preferred over $M_{0,r}$? (ln B > 0) (****)
- Yes → $M_{0,r}$ = $M_{1,r}$: start again
- No → M_1 = the pruned $M_{1,r}$

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moment detected? (*) 0 outside the 68% HPD of an amplitude
- Yes → $M_{1,r}$ = $M_{0,r}$ + the detected family with the highest residual evidence (**); full fit (all parameters free), pruned (***)
- No → is r of $M_{0,r}$ compatible with zero at 3σ?

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- Yes → $M_{0,r}$ = $M_{1,r}$: start again
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is $M_{1,0}$ (r = 0) preferred over $M_{0,r}$? (ln B > 0) (****)
- Yes → M_1 to the profile scheme
- No → detected

Res_r0 on the residuals of $M_{1,0}$: add to M_1 the family with the highest residual evidence (no pruning)

is r of $M_{0,r}$ compatible with zero at 3σ?
- none → **Res_r0** on the residuals of $M_{1,0}$: add to M_1 the family with the highest residual evidence (no pruning)
- Yes → upper limit
- No → r detection

upper limit / r detection

Bayesian result: always the posterior of r of $M_{0,r}$

then the profile scheme (with $M_{0,r}$ and M_1)

B · Profile scheme: every model at its own r

M* = $M_{0,r}$, the final model of the tree
candidates c = M* + every combination of the families added in the last Res_r0 (M_1 included), respecting their requirements (ds1 only with s)

$M_{0,r}$ and M_1: the lower and the upper end of the moments; the same candidates whether $M_{1,0}$ was preferred or not

profiles in r
M* → r* = its minimum
every candidate c → r^*_c = its minimum

profile: r fixed on a grid, all the other parameters re-fitted (ML)

compare all the models, each at its own r
M* at r*, every candidate c at r^*_c: local evidence (****) of each (its moment spectra free; β_d, β_s free in all, T_d only with dT)

final model = the one with the highest evidence
(M* itself if no candidate beats it)

profile of the final model in r
r_min = its minimum (r* or r^*_c); posterior of r = $\exp(-\Delta\chi^2/2)$ × prior of the tree

other parameters
posterior conditional on r = r_min (emcee, r fixed, the rest free)

0 within the central 3σ interval of the posterior of r?
- Yes → **upper limit**: r_95 of the posterior with r ≥ 0
- No → **r detection**: median and 68% of the posterior of r

(*) all the amplitudes of the family are fitted (emcee); detected if 0 is outside the 68% HPD (highest posterior density interval) of at least one of them; only the amplitudes with 0 outside their HPD are kept
(**) residual evidence: ln Z of the residual fit of each detected family; it sets the order in which the families are tried (Res_wr) or picks the one added (Res_r0)
(***) amplitudes with 0 in their 68% HPD (posterior of the full fit) removed and the model refitted, until all are detected; the pruned model is kept if ln B(pruned vs full) > 0 and is the one compared with $M_{0,r}$
(****) ln B within 2σ of the threshold (A) or between the two best models (B): PolyChord repeated with other seeds, mean of ln Z, up to 3 runs per fit